

Project Design Phase
Solution Architecture

Date	June 2025
Team ID	LTVIP2025TMID55104
Project Name	ResolveNow: Your Platform for Online Complaints
Maximum Marks	4 Marks

Solution Architecture Overview:

The ResolveNow system follows a client-server-based architecture using the MERN stack (MongoDB, Express.js, React.js, Node.js). The architecture bridges user interface requirements with backend logic and database storage, ensuring real-time communication, efficient complaint tracking, and modular code management.

Architecture Layers:

1. Frontend (Client Layer):

- **Technology:** React.js with Bootstrap & Material UI
- **Role:**
 - User-facing interface for complaint registration and tracking
 - Role-based access (Admin, User, Agent)
 - Axios used for API calls
 - Real-time dark/light theme toggle
 - Chat UI for interaction with agents

2. Backend (Application Layer):

- **Technology:** Node.js with Express.js
- **Role:**
 - Handles REST API endpoints
 - Authentication and session management
 - Complaint routing logic
 - Connects frontend to database

3. Database (Storage Layer):

- **Technology:** MongoDB Atlas (Cloud-hosted NoSQL DB)
- **Role:**

- Stores user profiles, complaints, status updates, chat history, and agent/admin data
- Document-based structure for flexibility and scalability

4. Optional Integration:

- Socket.io (For Real-Time Messaging)
- Email/SMS Gateway (For notifications)

Data Flow Overview:

1. **User Sign Up/Login:** User credentials are sent from frontend to backend, verified, and stored in MongoDB.
2. **Complaint Submission:** User submits a complaint, which is stored in the database and visible to admins.
3. **Assignment:** Admin assigns complaint to agent based on workload.
4. **Chat:** User and agent communicate in real-time (via chat module).
5. **Tracking:** Complaint status is updated by agents/admins and reflected on user dashboard.

Architecture Diagram:

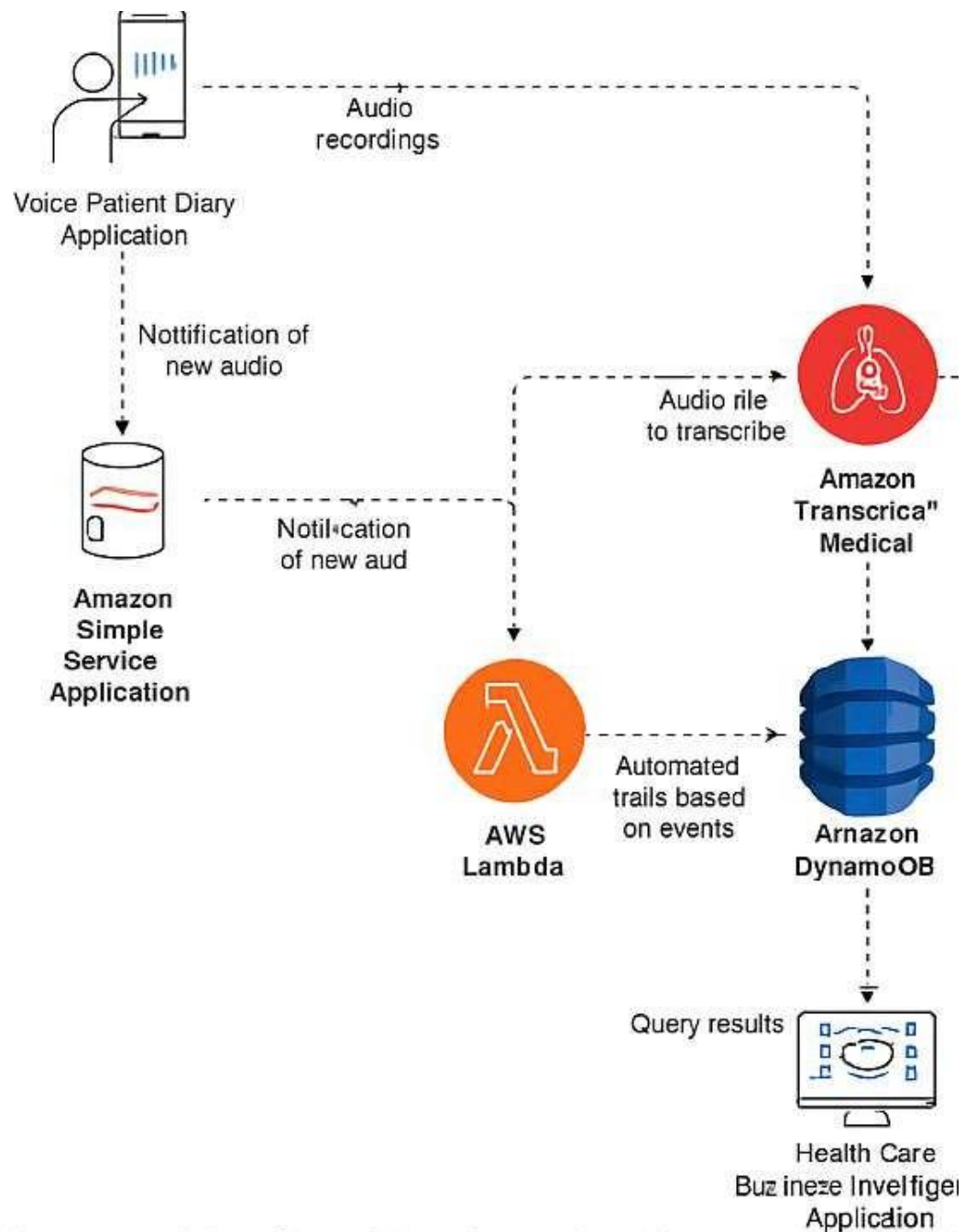


Figure 1: Architecture and data flow of the voice patient-diary sample application